

Varshini Elangovan

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EDUCATION

Carnegie Mellon University (CMU)

Master of Science in Machine Learning

Expected: December 2026

Pittsburgh, PA

Anna University, College of Engineering, Guindy

Bachelor of Engineering (Honors) in Computer Science and Engineering

June 2022

Chennai, India

WORK EXPERIENCE

Google

Software Engineer III

Hyderabad, India

November 2024 - July 2025

Software Engineer II

July 2022 - October 2024

- Drove risk strategy design for Google Wallet via PIX in Brazil, developing mitigation frameworks to reduce transaction losses across P2P and P2M payment flows for the August 2024 launch.
- Designed and deployed feature engineering pipelines in Java and rule-based decision policies in Python integrating ML model outputs for real time transaction risk assessment across Google Pay US and Wallet Brazil.
- Owned design, implementation, and validation of payment risk infrastructure extensions for recipients external to Google, including entity representation, control propagation, and downstream enforcement across the transaction pipeline.
- Engineered ML pipelines for delinquency fraud detection on Stored Value transactions, spanning data simulation, GBDT model training in TensorFlow and offline evaluation against production baselines.
- Built real-time monitoring dashboards and operational runbooks for regulatory compliance in Brazil. Collaborated with Trust & Safety and Product teams on post-launch metrics, thresholds and customer impact.

RESEARCH EXPERIENCE

Forge Lab, Carnegie Mellon University (Prof. Virginia Smith)

Pittsburgh, PA

Graduate Student Researcher (LLM Safety and Alignment)

May 2026 – Present

- Developing post-training interventions for LLM alignment, focused on identifying and suppressing unsafe persona-driven behaviours through on-policy consistency training and RL.
- Conducting a user study for [Safety Nudges](#), a real-time Chrome extension that audits chatbot conversations and surfaces contextual warnings for overconfidence, sycophancy, anthropomorphization and unsafe responses.

Human Sensing Lab, Carnegie Mellon University (Prof. Fernando De la Torre)

Pittsburgh, PA

Graduate Student Researcher (Multimodal Models)

October 2025 – Present

- Developed geometry-guided visual token pruning methods for multi-view 3D Visual Question Answering with 2D Vision-Language models, reducing inference cost by 60%, FLOPs by 88%, and KV-cache usage by 86% while maintaining comparable benchmark accuracy (in collaboration with Meta Reality Labs). Preprint coming soon.
- Working on safety benchmarking for Vision-Language Action models (in collaboration with Fujitsu Research).

Indian Institute of Technology (IIT), Kharagpur

Remote

Deep Learning Intern

May 2022 - September 2022

- Trained semantic segmentation models in PyTorch for depth estimation from simulated colonoscopy images as part of the KLIV group and studied effective post-processing techniques. Presented at the MICCAI 2022 Endoscopic Vision Challenge.
- **Publication: SimCol3D - 3D reconstruction during colonoscopy challenge**, A. Rau, S. Bano, Y. Jin, Varshini Elangovan, et al., Medical Image Analysis, 2024. doi: [10.1016/j.media.2024.103195](https://doi.org/10.1016/j.media.2024.103195)

PROJECTS

Watch & Learn: Teacher-Student Distillation for Robotic Arm Tasks ([GitHub](#))

November 2025 - December 2025

- Developed a vision-only student agent from the distillation of a full-state teacher policy for camera-based robotic manipulation on MetaWorld tasks. Achieved 90% of full-state teacher performance via teacher-guided RL with imitation learning.

Environmental ML: Modelling Sequential Disaster Cascades ([GitHub](#))

January 2026 – April 2026

- Built an end-to-end disaster cascade prediction pipeline on 15 years of NOAA Storm Events data, using multilabel classification, weather embedded neural networks and graph neural networks to model rare secondary hazards and spatial cascade patterns.

SKILLS

- **Languages:** Python, Java, C/C++, SQL, JavaScript
- **ML/Deep Learning:** PyTorch, TensorFlow, Keras, NumPy, Pandas, scikit-learn, Hugging Face, OpenCV, Gymnasium
- **Developer Tools:** Weights & Biases, Docker, Git, FastAPI
- **Research Interests:** VLMs, LLMs, post-training, RL, multimodal reasoning, technical AI safety